

Introduction

In the midst of global health emergencies like the COVID-19 pandemic, speed plays a crucial role. The urgent demand for fast-acting, effective therapeutics has pushed researchers to explore rapid-response solutions, one of which is aptamers. Aptamers are short, single-stranded DNA or RNA molecules (10-100 nucleotides in length) that are able to bind and neutralize targets with a high degree of specificity and affinity. While these molecules are specific, the complexity and sheer scale of the human biological landscape leaves room for uncertainty. Even well-designed aptamers may still produce unpredictable off-target effects or interactions. This uncertainty underscores the importance of the parallel development of countermeasures known as anti-aptamers, also known as “antidotes”, to ensure safety and control during the application of aptamers.

An interesting example of aptamer-based medicine is CoV2-6C3, a circular bivalent aptamer developed by Sun et al. in their paper “Aptamer Blocking Strategy Inhibits SARS-CoV-2 Virus Infection.” It was engineered to disable the interaction between the receptor-binding domain of the SARS-CoV-2 spike protein (S_{RBD}) and human angiotensin-converting enzyme 2 (ACE2) receptor based on previous research on SARS-CoV-2 neutralizing antibodies that perform the identical task. Aptamers offer numerous advantages over traditional antibody-based therapies. Their smaller size enables respiratory delivery and their high thermal stability and modifiable/personalizable nature make them adaptable to diverse therapeutic needs. Notably, aptamers are also less susceptible to antibody-dependent enhancement (ADE), a phenomenon where certain antibodies inadvertently facilitate viral entry into host cells, worsening the infection. These combined benefits make aptamers a compelling alternative. The results from Sun et al. concluded that CoV2-6C3 demonstrates a moderate binding affinity ($K_d = 44.78$ nM) and moderate viral inhibition ($IC_{50} = 446.68$ nM).

Despite the clear specificity and inhibiting power, the biological complexity of human biology raises the possibility of off-target effects. Therefore, we chose CoV2-6C3 for the design of an antidote as a proof-of-concept to demonstrate that aptamer activity can be reversed in the event of any off-target side-effects ensuring greater precision in a therapeutic application.

Methods

We first predicted the secondary structure of the focus aptamer, CoV2-6C3, by inputting the RNA sequence into mfold and RNAfold, two commonly used tools for RNA secondary structure prediction. These programs generate dot-bracket (dotb) notation outputs, thermodynamic details, and structure plots, allowing us to identify possible binding regions, such as loops, which are often involved in target recognition.

Using Python, we then derived the Watson-Crick complement of the aptamer's predicted binding site(s). Specifically, we focused on single-stranded loop regions due to their high accessibility and biological relevance. Loop-targeted antidotes are the most commonly used approach, as loops and bulges are exposed/unpaired and frequently comprise the aptamer's active or binding site. In order to increase affinity and stability, we also explored the possibility of multi-loop or bivalent designs, wherein a single antidote spans two adjacent or nonadjacent binding sites. These bivalent antidotes can achieve stronger and more stable binding due to multivalent interactions, as demonstrated in applications such as the circular aptamer cb-CoV2-6C3. However, these often require long oligonucleotides, which are harder to synthesize and more prone to degradation. They also impose thermodynamic/structural penalties as they require alignment across far binding zones. Thus, we opted to forego antidotes that bind across three or more aptamer sites due to unrealistic complexity.

Using the package NUPACK, we decided to simulate the aptamer-antidote complexes. Each antidote was evaluated by pairing it with the aptamer CoV2-6C3 in a 1:1 molar ratio (1 μ M each) within a simulated tube, representing a reaction system. We used NUPACK's Model class with the material parameter set to 'dna', as our antidotes were designed as DNA oligonucleotides targeting the RNA aptamer. We were then able to assess the binding strength (ΔG), structural favorability (MFE), and likelihood of complex formation under equilibrium conditions (concentration), allowing us to see the ranking of antidote effectiveness based on energetics and stability.

To further validate the physical interaction potential between the aptamer and its antidotes, we then performed 3D docking simulations using the HDock web server. First, we converted each aptamer and antidote sequence into 3D structures using a RNAComposer which outputs PDB coordinates. These structures were submitted to HDock for rigid-body docking. In each docking job, the aptamer was assigned as Molecule 1 and the antidote as Molecule 2. HDock computed docking scores based on a hybrid knowledge-based and energy-based scoring function, which includes van der Waals interactions,

electrostatics, and desolvation energy. Lower docking scores indicated more favorable binding conformations.

Results

Aptamer Analysis

To identify potential binding regions, we first analyzed the CoV2-6C3 aptamer using mfold and RNAfold (Figure 1). Both tools produced similar secondary structures in dot-bracket notation, with predicted ΔG values of -4.70 kcal/mol (Structure 1) and -3.90 kcal/mol (Structure 2), respectively. The minimum free energy structures revealed several single-stranded loop regions, which are unpaired and thus accessible to complementary binding. These loops served as the basis for targeted antidote design, as demonstrated in the Python file attached.

Sequence: CGCAGCACCCAAGAACAAGGACTGCTTAGGATTGCGATAGGTTCGG

Dot-Bracket: .(((....((.....)).)))).....(((.....))).

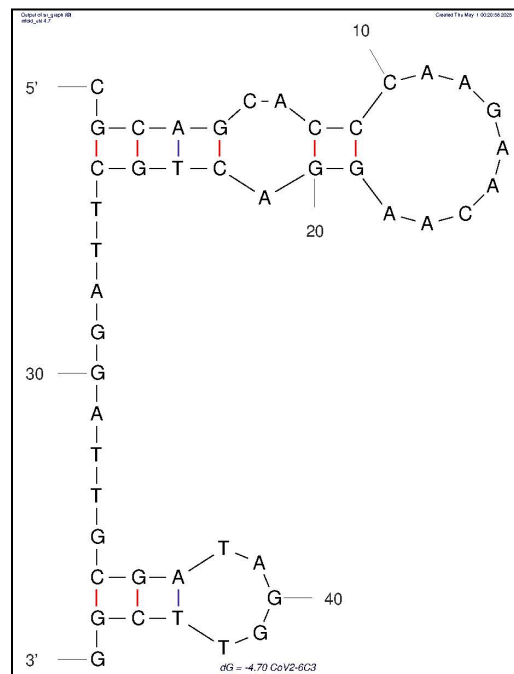


Figure 1: Secondary structure of the CoV2-6C3 RNA aptamer predicted using mfold. Loops and bulges, indicated by unpaired regions in dot-bracket notation, were selected as binding targets for antidotes.

Computational Binding

We then modeled the hybridization of CoV2-6C3 with multiple antidote candidates using NUPACK. Each complex was assessed for thermodynamic stability and binding likelihood using three key metrics: ΔG (binding favorability), MFE (structural stability), and equilibrium concentration (duplex formation probability).

Antidote Sequence	ΔG (kcal/mol)	MFE (kcal/mol)	Equilibrium Concentration (M)
TTGTTCTTG	-15.024539	-13.619834	3.693774e-08
CAATCCTAA	-18.544689	-17.913565	7.395034e-07
ACCTA	-12.868473	-11.345464	1.201795e-09
CAATCCTAATTTTTTGTTCCTTG	-20.561631	-19.380640	8.916840e-07
ACCTATTTTTTGTTCCTTG	-15.889857	-14.231894	8.877600e-08
ACCTATTTTCAATCCTAA	-18.918916	-17.935560	7.426921e-07

Table 1: Thermodynamic properties of aptamer–antidote complexes via NUPACK

From the paper, the binding affinity/ dissociation constant of the engineered aptamer is 44.78 nM.

$$K_d = 44.78 \text{ nM}$$

$$\Delta G = -RT \ln(K_d)$$

$$\Delta G = -(8.314 \text{ J/mol} \cdot \text{K} \cdot T) (298 \text{ K}) \ln(44.78 \text{ nM} \cdot 10^{-9} \text{ M/nM})$$

$$\Delta G = -41\,924 \text{ J/mol}$$

$$\Delta G = -41.9 \text{ kJ/mol}$$

Given that 1 kcal = 4.184 kJ,

$$\Delta G = -10.0 \text{ kcal/mol}$$

From this rough calculation, we see that the binding of the various antidotes to the aptamer has similar or better energies as the binding of the aptamers to the protein binding site. This indicates that the designed aptamers would likely bind to and inhibit the aptamer competitively.

Docking Simulation

To assess whether antidote binding could realistically disrupt aptamer–target interactions, we selected the antidote with the lowest ΔG (CAATCCTAATTTTTGTTCTTG), for 3D docking analysis alongside the aptamer and the SARS-CoV-2 spike protein. We used HDOCK to simulate docking between CoV2-6C3 and the SARS-CoV-2 spike RBD (Chain A), and CoV2-6C3 and the chosen antidote. Loop regions of the aptamer were annotated as active residues to signify docking toward biologically relevant binding sites. Antidotes were left fully flexible during the simulation.

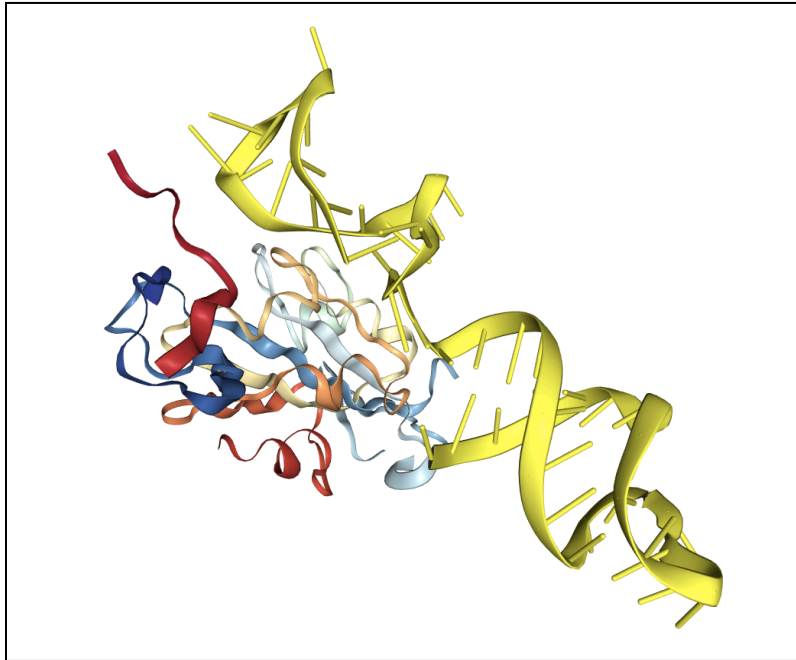


Figure 2: 3D Docking of CoV2-6C3 Aptamer (yellow) with Chain A of the Open Conformation RBD Region of the S protein of SARS-CoV-2 (rainbow).

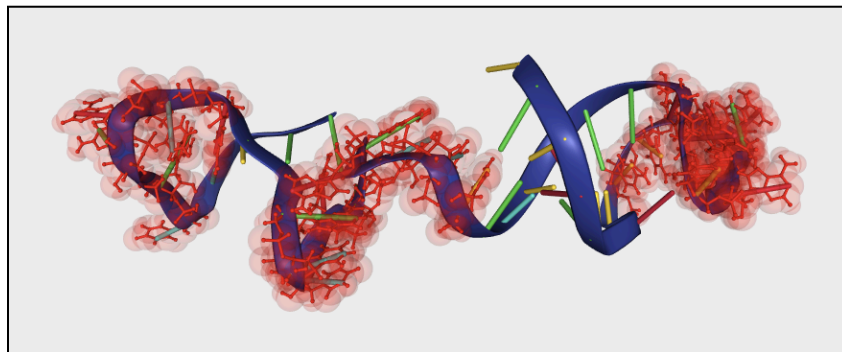


Figure 3: Secondary Structure of the CoV2-6C3 RNA Aptamer With Loop Residues Highlighted in Red.

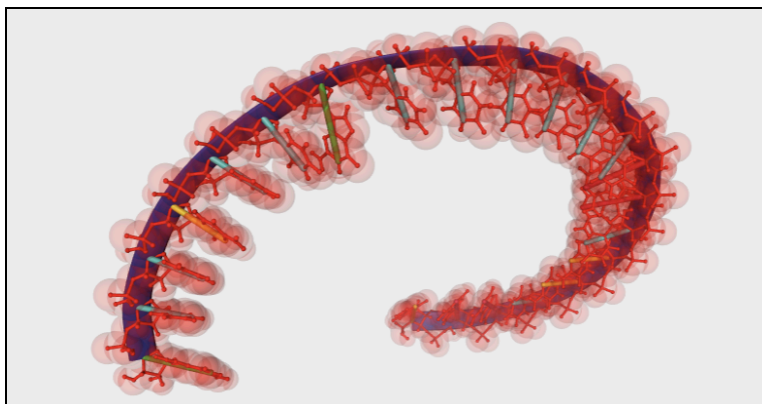


Figure 4: Secondary Structure of Antidote CAATCCTAATTTTTGTTCTTG with Loop Residues Highlighted in Red (from HADDOCK)

The docking score is calculated using HDOCK's iterative scoring functions (ITScorePP or ITScorePR). A more negative docking score suggests a higher likelihood of binding, though the score itself does not directly correspond to experimental binding affinity. HDOCK also provides a confidence score, derived from an empirical logistic function based on docking score values. Confidence scores closer to 1.0 indicate a higher probability of molecular interaction: scores above 0.7 suggest strong binding potential, scores between 0.5 and 0.7 imply possible interaction, and scores below 0.5 indicate a low likelihood of binding.

Table 2: Docking Metrics for Aptamer–Protein and Aptamer–Antidote Complexes

	Protein-Aptamer Complex	Aptamer-Antidote Complex
Docking Score	-255.18	-224.21
Confidence Score	0.8913	0.8152

Discussion

Our results support the hypothesis that rationally designed DNA or RNA-based antidotes can effectively bind to the CoV2-6C3 RNA aptamer and potentially neutralize its activity. The candidate antidote demonstrated favorable thermodynamic profiles, with low ΔG and MFE values and high equilibrium complex formation in NUPACK simulations. These findings suggest that short loop-complementary sequences that target structurally exposed regions can lead to antidotes with strong hybridization capacity. The longer bivalent antidote performed the best indicating that multivalent designs likely enhance stability, but may prove more difficult to synthesize and itself lead to adverse, off-target effects. Although untested, from literature review and a biochemical basis, we assume that by occupying the active loop regions of the aptamer, antidote binding would sterically or conformationally interfere with aptamer-target interactions. This is known as competitive inhibition. If the antidote is not sterically large or conformationally powerful enough for this task, modifications could likely be made to the antidote that attach large motifs which are even more disruptive.

Although we ran into issues while attempting docking (as this technique was not a focus of the course and not taught) based on the unoptimized results we recovered, we could somewhat conclude that these antidotes dock to the aptamer with at least a similar degree of intensity as the aptamer docks to the protein. Thus, these antidotes would function in a competitive inhibition model. Unfortunately, with computational models many areas to ensure a prediction that functions similarly to in vitro/in vivo are still lacking. To name a few, our models do not consider RNase activity, in vivo degradation, cellular uptake, or aptamer circularization effects (as in cb-CoV2-6C3). Moreover, while HDock provides structural insight, it remains a coarse approximation compared to molecular dynamics simulations or in vitro validation.

Future work should explore more extensive computational docking procedures (than the ones used here) and more far-reaching ways to generate various antidotes. The more testing that can be achieved computationally, the more stable and effective antidotes are likely to be discovered. Translating computational predictions into clinically viable aptamer-antidote systems is extremely complex and tricky, requiring much more work than shown here.

Conclusion

This short project demonstrates the feasibility of designing RNA/DNA-based antidotes to neutralize aptamers like CoV2-6C3, through structure-based modeling. Using secondary structure prediction, thermodynamic simulations, and docking analyses, we identified several candidate antidotes with potential for effective binding and inhibitory properties. This project offers an (extremely) rough proof-of-concept for embedding a control mechanism into aptamer-based therapies to serve as a safeguard against potential off-target effects. Antidotes represent an important step toward developing reversible, tunable therapeutics.

References

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